



**VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES - TECHNICAL
CAMPUS Grade A++ Accredited Institution by NAAC**

NBA Accredited for MCA Programme; Recognized under Section 2(f) by UGC;
Affiliated to GGSIP University, Delhi; Recognized by Bar Council of India and AICTE
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SCHOOL OF ENGINEERING & TECHNOLOGY

ASSIGNMENT 2 (UNIT-III & IV)				
SUBJECT: SUPERVISED AND DEEP LEARNING		SUBJECT CODE: ML 463 T		SEM: VII BRANCH: CSE
S. NO	QUESTION	CO BL		MARKS
1	What are the main gates in LSTM and what are their tasks?	2	1,2,3,4	5
2	What is transformer architecture, and why is it widely used in natural language processing tasks? What is Self-attention Mechanism?	2	1,2,3,4,5	5
3	You have an input image of size $32 \times 32 \times 3$ (height $\times$ width $\times$ channels). You apply a convolution layer with: • 8 filters of size 5×5, • stride = 1, • padding = 0 (i.e., valid padding). Find: 1. The output feature map size (height $\times$ width $\times$ depth). 2. The number of trainable parameters in this layer. Why do we use ReLU activation in CNNs instead of sigmoid or tanh?	2	1,2,3	5
4	How is an autoencoder different from PCA ? What is the difference between inductive , transductive , and unsupervised transfer learning?	2	1,2,3	5